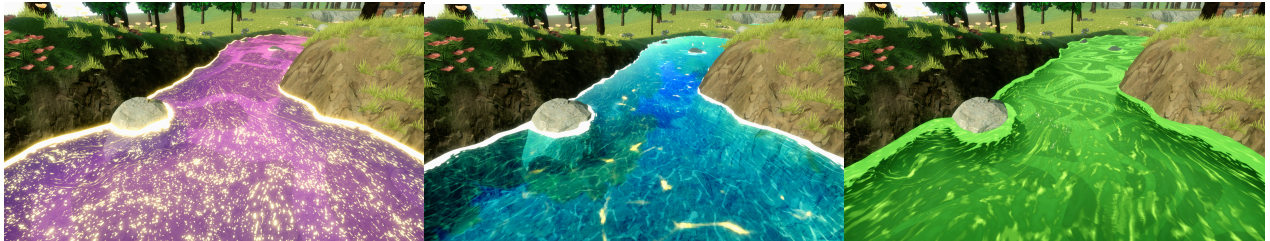


AQUIS Pro - Official Documentation

1. Introduction

First of all, thank you for purchasing this asset. Secondly, feel free to contact me with any questions, bug reports, or feature requests you'd like to see in future versions. **AQUIS Pro** is a professional solution for generating rivers and other water bodies in Unity URP, supporting both orthographic and perspective cameras. This asset combines a powerful spline-based mesh generator with a highly customizable Water Shader, as you can see in the images.

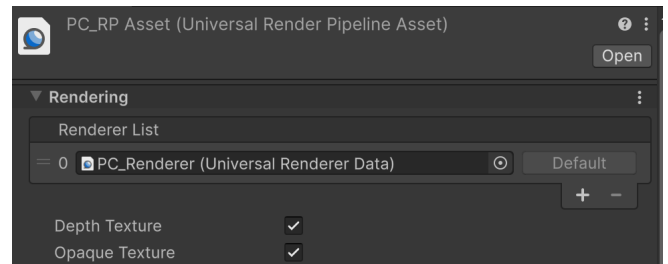
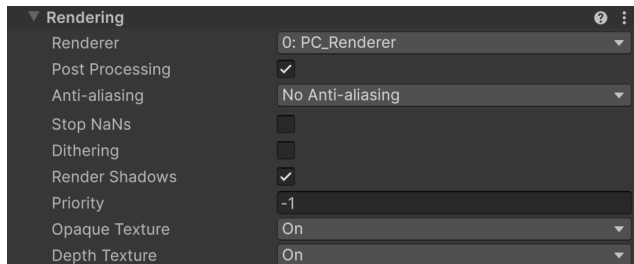
It also includes an automatic flow map generator that allows you to paint directly onto the mesh to configure the river's flow, among other features we will explore in depth throughout this documentation. Just have fun with it. I created it with a lot of hard work and learning so that any developer like me can enjoy it.



2. Fast Set Up

First and foremost, this asset only works with the Universal Render Pipeline (URP). Compatibility is not guaranteed with versions older than Unity 6. Both Depth Texture and Opaque Texture must be enabled in your URP Render Pipeline Asset to see some of the water features. These parameters can also be forced on the Main Camera.

Remember that the asset is located inside [AureDevGames/AQUISPro](#).



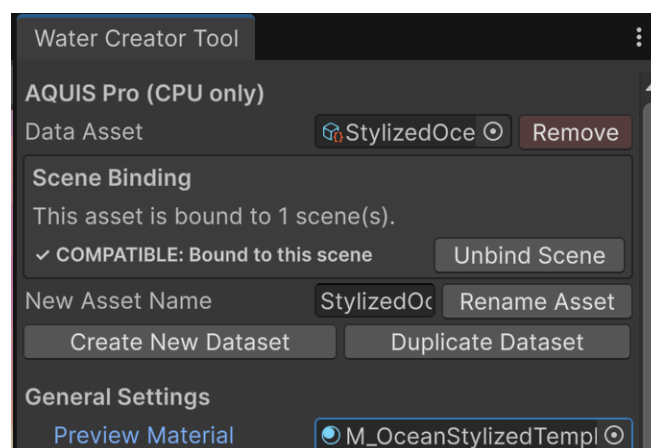
- The Quickest Way

The fastest way to generate water is by dragging a prefab from the Prefabs/Water folder directly into your scene. However, these objects are mostly intended as template examples and might not perfectly fit every need, though they can do the trick in some cases.

- The Best Way

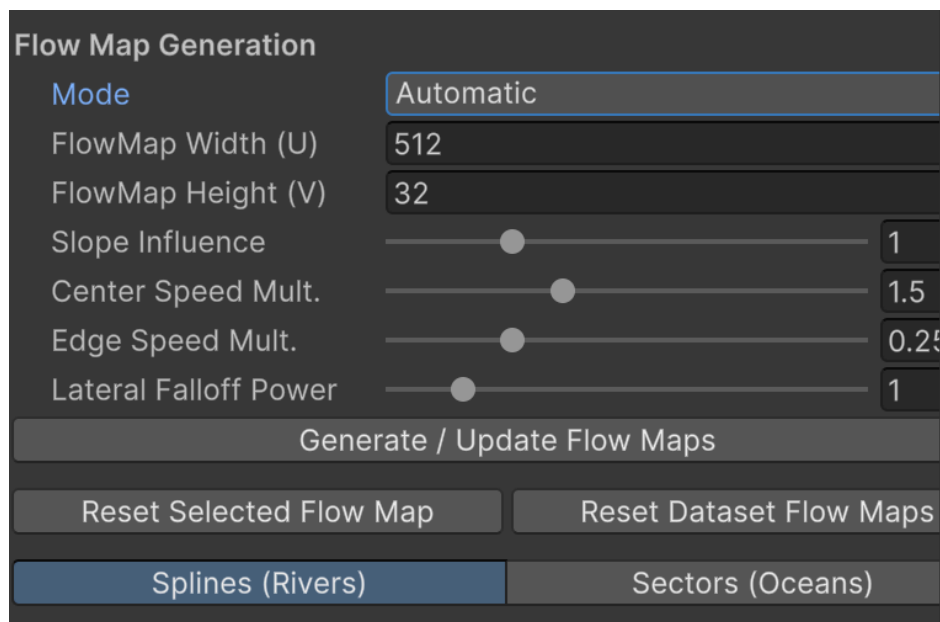
If you bought this asset, I imagine you want to tinker with it:

1. Go to the **Tools** tab and select **Aquis PRO**.
2. A window called **Water Creator Tool** will open.
3. Click **Create New Dataset** to start a water body from scratch, or **Duplicate Asset** to use one of the provided examples as a base. You will need to select that asset in the **Data Asset** field. You can also rename it using **Rename Asset**, which is especially useful if you duplicated it. After that, don't forget to click **Bind Scene** so the asset always exists in that specific scene (click **Unbind Scene** if you want to remove it). Finally, assign a material from the Water Shader (like any found in [Preset Materials/Rivers](#) or [Preset Materials/Oceans](#)).



Now, the workflow branches depending on whether you want to create a river (spline-based) or a standard ocean (sector-based). Follow path **A** for a river, or path **B** for an ocean.

4a. Splines (Rivers) Select the **Splines (Rivers)** menu. Control the tessellation level with the **Tessellation Level** slider. Also, control the **U and V Tiling Multiplier** and assign an aspect ratio similar to the space the river will occupy. If you want the mesh to adapt to the shape of your terrain, enable the **Snap Mesh to Riverbed** option and select the target object's layer as the collision mask. This option is highly recommended to make the most of the automatic flow map generation. You can click **Enable Reflection Probes** to automatically generate reflection probes at intervals, which helps shape the reflections later. They are generated in Realtime by default, but baking them is recommended. Once that's done, move on to creating the splines. You can place them by clicking **Add Spline** and adjusting the handles, but using **Draw Mode** is highly recommended. In this mode, you choose a target layer and a Y offset, allowing you to easily place splines wherever you want, and they will adjust automatically. By holding click, you can modify the curvature before placing it, as well as easily undo if necessary. When you select a spline segment, **Width** and **Flow Speed** can be modified, which will smoothly blend with the rest of the segments. For flow maps, click **Generate/Update Flow Maps** to see what it does. By enabling **Debug Flow Map** on the material, you will be able to see how a flow map is



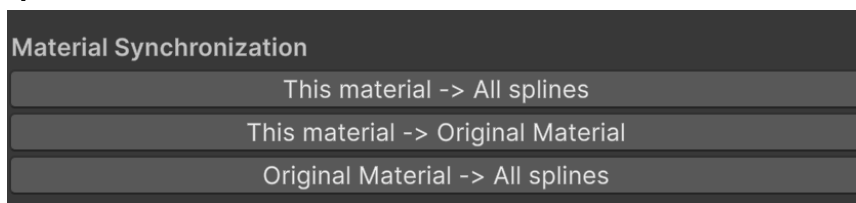
generated for each segment and assigned to the material. The Red and Green channels determine the direction, while the Blue channel determines the speed. You can tweak various self-explanatory settings (hover over them for tooltips) to modify your automatic flow map generation. However, if you need more specific control—like changing the flow around a rock—you should use the **Custom Mode** in Flow Map Generation. In Custom mode, you can choose the brush size and speed. By clicking **Click To Enable Paint Mode**, you can paint the desired flow map directly onto the spline in the editor, saving your brush stroke direction as the water direction.

4b. Sectors (Oceans) Select the **Sectors (Oceans)** menu. When selected, many river-specific options will disappear. Keep the **U and V Tiling Multiplier** roughly proportional to the width and length of your ocean. Click **Add New Water Sector** to start editing. A water plane will appear. In the menu, you can control the plane's position, width, length, vertex density, and how it is divided into X and Z sectors. Sectors are used to divide the plane and take better advantage of **Adaptive Tessellation** (explained later).

5. Select the created parent object, named by default **WaterCreator_Previews_YourAssetName**. In the inspector, you will see the **Water Waves** script, a handler to easily modify the shader and its wave settings. Choose the parameters you find suitable for your water body's waves, as well as the wave direction and randomness level. You can individually tweak the four Gerstner waves combined in the operation. Next, you can adjust the **Foam**, which generates from the highest points of the waves. Hovering over a property will give you a brief explanation. When you finish configuring this, click **Save configuration to data set** to save these properties. You can also easily load them later with the **Load from data set** button.

6. If you have a high vertex density, you should optimize it by enabling **Adaptive Tessellation**. This dynamically changes the tessellation level of the tool's mesh depending on the screen space it occupies. Adapt the three available quality levels to your needs, it's common to set the lowest level to a tessellation of 1 so the water becomes a simple plane in the distance.

7. If you are working on sectors, just modify the material of any sector of your data to see everything change, but for the rivers you have three options. Every spline of the river has a different material due to the flow map support, so if you want the changes you are doing in one spline to be in the others, you have these buttons. **This material to All splines** synchronizes the material of the spline selected with the rest. **This material to Original Material** changes the "father material", the field Preview Material of the data, with the current material of the selected spline. Finally, you can override every spline with the preview material using **Original Material to All splines**.



8. You should tweak these parameters alongside the water material until you are satisfied with the results. Once you achieve a look you like, click **Export selected Data Set as Prefab**. You can always overwrite it later by clicking the **Override** button.

3. Water Shader Properties

The second major part of the package is a layered water shader with a clear pipeline that you can easily explore in Shader Graph. I will explain the different layers and how they are applied in order. The shader properties always include a short tooltip explanation if you hover your mouse over them for a moment. With that said, let's proceed.

3.1. Depth, Fresnel and Caves

In the first calculation of the shader, we determine which parts of the objects are submerged in water and which are not. For the submerged parts, we apply a specific refraction and color. To this, we add a Fresnel effect that occurs when the water aligns with the horizon in your view, giving the sensation of reflecting a strong color. In this step, we also add the "Caves" feature—a texture located below everything else to provide a sense of depth, particularly for a more stylized or cartoonish artistic purpose.

3.2. Primary Foam, Secondary Foam and Flow Map

Here, we decide which textures will pan across the water material. If we want standard panning using the Spline or Sector UVs, we just need to disable the **FlowMap?** toggle on the material. When disabled, the Primary & Secondary Foam textures are used directly. If enabled, those two properties are ignored, and the texture will use the flow map generated on the spline to apply the desired direction and speed. This opens up a world of possibilities; you can experiment with different textures to see the resulting effects.

3.3. Illumination

Next, we add the generation of halos and sparkles that depend on the scene's lighting. This also supports point lights and their shadows.

3.4. Waves & Foam

We continue the pipeline by calculating the vertex positions using Gerstner Waves. Here, we also calculate where the crest foam should appear and add it to the fragment shader pipeline.

3.5. Water Reflections

Reflections are added using the information from nearby reflection probes, or by using the skybox texture for greater simplicity.

3.6. Intersection Foam

We calculate where objects intersect with the water using a depth function similar to the one in the first step. Based on that intersection, a foam texture is created that evolves and adapts to the water's movement.

3.7. Fog

Near the end, the color from Unity's native Fog is added to the water shader.

3.8. Caustics

Caustics simulate the scattered light that passes through the water and hits the underwater floor. The depth texture is used to measure the depth and perfectly map the texture to the volume. In this version, caustics are only visible when looking through the water surface.

3.9. Color adjustments

To improve the color palette settings and artistic feel, sliders have been added to control the final HSV (Hue, Saturation, Value) of the river, allowing for final adjustments.

4. Coming Up

There are some features that couldn't be implemented yet but will be coming in future updates.

- **4.1. Planar reflections:** I really want to implement this reflection method. Although more performance-heavy, it is simpler and more precise for the developer to adjust.
- **4.2. Water decals:** I want to implement an easy way to add decals on top of the water using the existing tool, perfect for things like algae or dirt on the surface.
- **4.3. Waterfalls:** I am aware of how necessary waterfalls are for creating elevation changes in a river. That's why I've reserved this feature to ensure it integrates properly with the existing elevation logic used for flow map calculation.
- **4.4. Shorelines on the coast:** This is definitely a high priority. I want the beach intersections to look great by utilizing different techniques.
- **4.5. Screen space caustics & Underwater:** It might be useful for many users to have a fully developed Underwater effect with its respective caustics, without needing to view them solely through the top water surface.
- **4.6. Signed Fields for intersection foam:** I have researched some methods to make intersection foam look better from profile/side angles, which is the current flaw when calculating it via depth. Although it's more expensive computationally, having the option to calculate it with this method could increase visual fidelity.
- **4.7. GPU use for performance improvements:** I would like to add a mode where the geometry and waves are controlled by gpu to remove CPU weight.
- **4.X. Anything you can think of:** That's right! Don't be shy—contact me about any update you might need for your project or any feature you'd like to see, and I'll see what I can do 😊

5. Contact

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